# Evaluating Multilingual Reliability of Large Language Models in Scientific Literature Analysis

*A Review of Cross-Lingual Factuality, Hallucination, and Benchmarking Approaches for LLM-Assisted Scientific Text Analysis*


## Abstract

Large language models (LLMs) are increasingly deployed to search, summarize, translate, and synthesize scientific literature, yet the overwhelming majority of evaluation evidence for these capabilities is drawn from English-language benchmarks and corpora. This paper reviews the current state of research on the multilingual reliability of LLMs when applied to scientific literature analysis, drawing on cross-lingual factuality studies, multilingual benchmarking efforts, and empirical evaluations of LLM-assisted literature review pipelines. We synthesize evidence showing that model accuracy, factual consistency, and hallucination rates degrade systematically as one moves from high-resource languages such as English and Mandarin toward mid- and low-resource languages, and that this degradation persists even in state-of-the-art proprietary and open models. We examine current approaches to mitigating these gaps, including retrieval augmentation, cross-lingual chain-of-thought prompting, multilingual instruction tuning, and multilayered quality-control pipelines for automated review generation, and we assess their demonstrated effectiveness and limitations. We further identify methodological, linguistic, and infrastructural research gaps, including the scarcity of non-English scientific corpora, the absence of standardized multilingual scientific-reliability benchmarks, and the risk that LLM-mediated literature synthesis could entrench English-centric knowledge hierarchies within global science. We conclude with directions for future research aimed at building more linguistically equitable and verifiably reliable LLM tools for scientific literature analysis.



### *Keywords*

*large language models; multilingual natural language processing; scientific literature analysis; cross-lingual consistency; hallucination; systematic review automation; low-resource languages; factuality evaluation*


## 1. Introduction

The volume of peer-reviewed scientific literature has grown to a point where manual review, screening, and synthesis of relevant publications increasingly exceed the practical capacity of individual researchers (Wu et al., 2025). In response, large language models (LLMs) such as GPT-4 (OpenAI, 2023) have been rapidly adopted as tools for literature search, abstract screening, data extraction, translation, and even full review composition (Scherbakov, Hubig, Jansari, Bakumenko, & Lenert, 2025). Systematic reviews of

this emerging literature report that GPT-based architectures dominate current review-automation projects, with the large majority of published pipelines built exclusively around English-language workflows (Scherbakov et al., 2025).

This English-centric orientation raises a critical but comparatively underexamined question: how reliable are LLMs when scientific literature analysis is conducted in, or extended to, languages other than English? Scientific knowledge is not confined to English-language publications. Regional journals in Chinese, Spanish, French, Japanese, Portuguese, German, Arabic, and dozens of other languages contain findings that are frequently underrepresented in English-trained corpora and in the pretraining data of general-purpose LLMs (Xuan et al., 2025). At the same time, researchers, clinicians, and policymakers around the world increasingly query LLMs about scientific findings in their native languages, expecting outputs that are as accurate and internally consistent as those generated in English. Evidence reviewed in this paper indicates that this expectation is not yet reliably met.

This paper addresses three interrelated questions. First, what does existing empirical evidence show about the reliability of LLMs across languages when performing scientific literature analysis tasks such as summarization, entity extraction, factual question answering, and review synthesis? Second, what techniques are currently being used to narrow the observed cross-lingual reliability gap, and how effective are they? Third, what methodological and infrastructural gaps remain, and what research directions could most plausibly close them? In addressing these questions, the paper synthesizes findings from multilingual benchmarking research (Hu et al., 2020; Xuan et al., 2025), cross-lingual factuality and hallucination studies (Qi, Fernández, & Bisazza, 2023; Li et al., 2024), and empirical evaluations of LLM-assisted literature review systems (Scherbakov et al., 2025; Tang, Duan, & Cai, 2025; Wu et al., 2025).

The remainder of the paper is organized as follows. Section 2 reviews the background literature on multilingual LLM development and on LLM-assisted scientific literature analysis. Section 3 presents the main thematic discussion of multilingual reliability, organized around factual consistency, hallucination, and resource asymmetry. Section 4 surveys current mitigation approaches. Section 5 discusses research challenges and limitations. Section 6 outlines research gaps and future directions, and Section 7 concludes.

## 2. Background and Related Work

### *2.1 The Rise of Multilingual Large Language Models*

Early transformer-based language models were predominantly monolingual or English-dominant, but subsequent efforts sought to extend coverage to dozens or hundreds of languages. The mT5 model extended the text-to-text transfer transformer architecture to 101 languages using the mC4 corpus (Xue et al., 2021), while the BigScience collaboration released BLOOM, a 176-billion-parameter open-access model trained on the ROOTS corpus spanning 46 natural languages and 13 programming languages, explicitly framed as a step toward democratizing access to large-scale multilingual modeling (BigScience Workshop et al., 2022). Proprietary systems such as GPT-4 (OpenAI, 2023) likewise claim broad multilingual competence, reporting strong performance when translated versions of the MMLU benchmark are evaluated across dozens of languages, although independent replications have generally found the reported gains to be uneven across the language spectrum.

To evaluate cross-lingual generalization systematically, Hu et al. (2020) introduced XTREME, a multi-task benchmark spanning 40 languages and 9 tasks, demonstrating that although models tested on English routinely approach human performance on many tasks, a substantial performance gap persists for cross-lingually transferred models, particularly on syntactic and sentence-retrieval tasks. More recent work extends this line of inquiry to instruction-following LLMs specifically: MMLU-ProX, a reasoning-focused multilingual benchmark covering 29 typologically diverse languages with parallel question sets, found that state-of-the-art LLMs achieved over 70% accuracy in English while performance for lower-resource languages such as Swahili fell to around 40%, with gaps of up to 24.3 percentage points between high- and low-resource languages evaluated on identical content (Xuan et al., 2025).

### *2.2 LLMs in Scientific Literature Analysis*

A parallel and increasingly active line of research examines the use of LLMs to automate components of the scientific literature review pipeline, including search, title/abstract screening, full-text screening, data extraction, and narrative synthesis. A large language model-assisted systematic review by Scherbakov et al. (2025), published in the Journal of the American Medical Informatics Association, retrieved 3,788 candidate articles and identified 172 eligible studies describing LLM-based review-automation projects, finding that ChatGPT and GPT-based architectures accounted for 73.2% of the systems studied, and that systematic reviews were the most frequently automated review type (68.6% of cases). The authors reported that automation quality remained uneven across pipeline stages, with numeric data extraction showing markedly lower accuracy than qualitative screening tasks.

Complementary evidence comes from Wu et al. (2025), who developed a fully automated review-generation pipeline evaluated on 343 articles across 35 topics in propane dehydrogenation catalysis, reporting that multilayered quality-control procedures reduced hallucination risk in generated citations to below 0.5% with 95% confidence when expert verification was applied. Similarly, a medRxiv preprint describing a Claude-API-based fully automated systematic review pipeline reported citation accuracy of 95.87% after applying controlled text-restriction strategies designed to constrain the model to verifiable source content. Tang, Duan, and Cai (2025) introduced a multidimensional evaluation framework for automated literature review composition and found that even advanced LLMs, evaluated on reference generation, abstract writing, and full review composition, continued to generate hallucinated references and showed inconsistent performance across academic disciplines.

Notably, nearly all of this literature-review automation research has been conducted, evaluated, and reported in English. The scoping review by the Discover Artificial Intelligence journal explicitly excluded non-English publications from its inclusion criteria when surveying LLM applications to biomedical text synthesis, illustrating a broader pattern in which the empirical evidence base for LLM-assisted literature analysis is itself linguistically narrow, even as claims about the general applicability of these tools are frequently made without qualification.

### *2.3 Cross-Lingual Factuality and Hallucination*

A separate but directly relevant body of work examines whether LLMs represent and retrieve factual knowledge consistently across languages. Qi, Fernández, and Bisazza (2023), in an Outstanding Paper Award-winning study at EMNLP 2023, proposed a Ranking-based Consistency (RankC) metric and found that increasing model size improved factual probing accuracy in most languages but did not improve cross-lingual consistency; moreover, when new factual associations were inserted into a model via editing in English, the new knowledge transferred only to languages with which English already exhibited high consistency scores. This finding has direct implications for scientific literature analysis, where new findings often first appear in English-language preprints or journals: LLMs may fail to propagate updated scientific knowledge into other languages even after being exposed to it.

Related work on multilingual confidence estimation similarly found that English exhibits notable dominance in model confidence calibration relative to other languages, and that prompting strategies aligned with the query language can partially close this gap (Li et al., 2024). Studies of hallucination detection likewise report that a general trend holds across the literature: outputs in English tend to be the most factually stable, with quality declining as language resource level decreases, a pattern attributed in part to asymmetric knowledge coverage in pretraining corpora and to the absence of an explicit cross-lingual semantic alignment objective in standard next-token-prediction pretraining.

## 3. Main Thematic and Technical Discussion: Dimensions of Multilingual Reliability in Scientific Contexts

### *3.1 Resource Asymmetry and Its Propagation into Scientific Tasks*

The dominant explanatory factor identified across the reviewed literature is resource asymmetry: the overwhelming majority of pretraining tokens available to large models originate from English and a small set of other high-resource languages, while thousands of the world's languages, including many with substantial scientific publishing traditions, are represented by orders of magnitude less data. Xuan et al. (2025) quantify this asymmetry empirically, showing consistent performance degradation from high-resource to lower-resource languages across a reasoning-focused, translation-verified benchmark. Because scientific literature analysis tasks such as terminology extraction, methodological classification, and claim verification depend on precise factual and technical understanding rather than surface fluency, this resource asymmetry does not merely produce awkward phrasing in low-resource-language outputs; it produces materially different factual content, including omitted findings, incorrect numerical values, and misattributed claims.

### *3.2 Cross-Lingual Factual Inconsistency*

A second and more insidious dimension of unreliability is internal inconsistency: an LLM may generate different, sometimes contradictory, factual claims about the same scientific question depending solely on the language in which it is queried. Qi et al. (2023) demonstrate that this is not simply a function of raw accuracy differences between languages but a distinct phenomenon of representational misalignment, in which the same underlying fact is encoded and retrieved differently depending on the query language. For scientific literature analysis specifically, this raises the possibility that a researcher querying an LLM about drug efficacy, statistical results, or methodological details in French may receive a materially different answer than a colleague posing an equivalent query in English, even when both are nominally asking about the same underlying publication or body of evidence. Cross-lingual and cross-modal factuality benchmarks report that this inconsistency problem is exacerbated further when queries combine multiple languages or modalities, indicating that the underlying alignment gap compounds rather than averages out under more complex, realistic query conditions.

### *3.3 Hallucination in Multilingual Literature Synthesis*

Hallucination, broadly defined as the generation of content that is fluent but factually ungrounded, has been identified as a fundamental and unresolved challenge that is particularly pronounced in multilingual settings, where limited training data increases the likelihood of inaccurate or misleading outputs. In the specific context of scientific literature review generation, Tang, Duan, and Cai (2025) found that even the

most advanced LLMs continued to produce hallucinated references, with performance varying meaningfully by academic discipline, underscoring that hallucination is not a uniformly distributed error but one shaped by domain-specific training data density. Retrieval-augmented generation (RAG) has demonstrated measurable effectiveness at reducing hallucination in English contexts, but has been shown to perform substantially worse in low-resource settings owing to unreliable retrieval indices and limited language coverage in the underlying retrieval corpora. Chain-of-thought prompting improves factuality in some cases but implicitly assumes that the model already possesses adequate internal knowledge in the target language, an assumption that does not hold for underrepresented languages and therefore limits the technique's usefulness precisely where it is most needed.

### *3.4 Domain-Specific Multilingual Gaps in Biomedical and Scientific NLP*

Biomedical and clinical natural language processing offers a particularly well-documented illustration of these dynamics. A medRxiv study on multilingual biomedical named entity recognition and entity linking found that concept-extraction tools for non-English biomedical texts, such as Dutch and Italian clinical notes, remain sparse and are often evaluated without the methodological rigor applied to English-language systems, a gap the authors attribute to the scarcity of annotated non-English biomedical corpora. Even when LLMs such as GPT-4o are applied in zero- and few-shot settings to multilingual clinical entity extraction, performance in non-English settings remains comparatively underexplored relative to English clinical NLP. Bilingual biomedical models such as Taiyi, fine-tuned on a curated collection of 140 biomedical text-mining datasets spanning 102 English and 38 Chinese datasets, demonstrate that targeted bilingual fine-tuning can outperform general-purpose LLMs on in-domain tasks, suggesting that architecture alone is insufficient and that curated multilingual domain corpora remain a binding constraint on reliability.

A broader survey of pretrained language models for processing scientific text similarly finds that scientific language models (SciLMs) have been developed for only a handful of non-English languages, predominantly Chinese, French, Japanese, and Spanish, and attributes the scarcity of coverage in other languages to the absence of large-scale, high-quality non-English scientific datasets, compounded by the practical difficulty of collecting and curating such datasets when the majority of research articles and reports are originally authored in English.

### *3.5 Safety, Confidence Calibration, and Downstream Risk*

Reliability failures in multilingual scientific contexts are compounded by uneven safety and confidence calibration across languages. Multilingual safety benchmarking work covering 45,000 entries across 12 languages found that safety and helpfulness outcomes vary significantly across domains and languages,

even among languages with comparable resource levels, indicating that resource level alone does not fully predict reliability and that language-specific idiosyncrasies in training data and alignment procedures also matter. Confidence estimation research similarly shows that models are frequently overconfident when producing incorrect answers in lower-resource languages relative to English, which is particularly consequential in scientific and clinical contexts, since overconfident but incorrect model outputs are harder for non-expert users to detect and are less likely to trigger appropriate verification behavior.

## 4. Current Approaches and Developments

### *4.1 Retrieval-Augmented and Knowledge-Grounded Generation*

Retrieval-augmented generation (RAG) is among the most widely adopted mitigation strategies, supplying the model with retrieved source passages at inference time rather than relying solely on parametric knowledge. Knowledge-graph-grounded approaches to multilingual hallucination evaluation report absolute improvements of approximately 0.12 to 0.36 points in semantic similarity scores when structured knowledge graph integration is used in place of vanilla question answering, across multiple languages and multiple models, indicating that grounding generation in externally verifiable structured knowledge can partially close the multilingual factuality gap. However, this benefit remains contingent on the availability of adequately populated knowledge graphs and retrieval corpora in the target language, which themselves exhibit the same resource asymmetry documented for pretraining corpora more broadly.

### *4.2 Cross-Lingual Chain-of-Thought and Knowledge Transfer*

A second family of approaches seeks to transfer reasoning capability from high-resource to low-resource languages at inference time. Adaptive multilingual chain-of-thought methods route intermediate reasoning steps through a high-resource pivot language before returning outputs in the target language, and have been reported to narrow, without fully closing, the performance gap between high- and low-resource languages while preserving cultural and linguistic nuance in final outputs. Cross-lingual knowledge-transfer methods designed specifically to mitigate hallucination in low-resource generation similarly report measurable reductions in context-related hallucination, defined as output that contradicts general world knowledge, by explicitly aligning retrieved or transferred knowledge representations across the source and target languages.

### *4.3 Multilingual Instruction Tuning and Domain-Specific Fine-Tuning*

Beyond inference-time interventions, targeted multilingual and domain-specific fine-tuning has shown consistent benefit. The Taiyi model, fine-tuned across 140 biomedical datasets spanning English and

Chinese using a two-stage supervised strategy, achieved superior performance relative to general-purpose LLMs on biomedical named entity recognition, relation extraction, text classification, and question answering, demonstrating that even bilingual (rather than fully massively multilingual) fine-tuning on carefully curated domain corpora can produce meaningful reliability gains for scientific subdomains where such corpora exist.

### *4.4 Multilayered Quality Control in Automated Review Pipelines*

At the application layer, several research teams have implemented multilayered, human-in-the-loop quality-control procedures for LLM-assisted literature review generation rather than relying on single-pass model outputs. Wu et al. (2025) report that layered verification, combining automated citation checking with expert review, reduced hallucination risk in a large-scale automated review-generation system to below 0.5%, while a Claude-API-based fully automated systematic review pipeline achieved a reported citation accuracy of 95.87% through controlled text-restriction strategies that constrain generation to verifiable source content. Multi-agent frameworks such as LatteReview further decompose the review process into modular agents responsible for discrete tasks, including title and abstract screening, relevance scoring, and structured data extraction, operating within orchestrated workflows that support iterative human feedback, an architecture intended to make errors easier to localize and correct than in monolithic single-model pipelines. To date, however, these quality-control frameworks have been developed, validated, and reported almost exclusively for English-language corpora, and it remains an open empirical question whether comparable hallucination-reduction rates are achievable when the same architectures are applied to non-English scientific literature.

## 5. Research Challenges and Limitations

### *5.1 Scarcity of Non-English Scientific Corpora and Benchmarks*

The single most frequently cited constraint across the reviewed literature is the scarcity of large-scale, high-quality non-English scientific text corpora suitable for both training and evaluation. This scarcity is self-reinforcing: because most scientific publishing infrastructure, indexing services, and annotation efforts have historically prioritized English-language content, the datasets required to train and rigorously benchmark multilingual scientific-reasoning capability in LLMs are themselves underdeveloped, which in turn limits the ability of researchers to even measure the scope of the reliability gap with precision.

### *5.2 Translation Artifacts and Semi-Automated Benchmark Construction*

Many existing multilingual benchmarks, including MMLU-ProX, are constructed via semi-automatic translation pipelines in which LLM-generated translations are subsequently reviewed by human

annotators for conceptual accuracy and terminological consistency (Xuan et al., 2025). While this hybrid approach improves scalability relative to fully manual translation, it introduces a methodological circularity risk: benchmarks intended to evaluate LLM multilingual reliability are themselves partly constructed using LLMs, and translation artifacts, particularly in technical or domain-specific scientific terminology, may be inconsistently caught by human review, potentially masking or distorting the true magnitude of cross-lingual performance gaps.

### *5.3 Confounding Between Language Resource Level and Domain Specificity*

Existing studies often struggle to disentangle general language-resource effects from domain-specific effects. A model may perform poorly on a scientific query in a given language either because the language itself is underrepresented in pretraining data or because the specific scientific subdomain is underrepresented regardless of language, or both simultaneously. Few of the studies reviewed here explicitly decompose these two sources of error, which limits the precision with which mitigation strategies, whether general multilingual fine-tuning or domain-specific fine-tuning, can be targeted and evaluated.

### *5.4 Absence of Standardized Multilingual Scientific-Reliability Metrics*

Whereas general-purpose multilingual benchmarks such as XTREME (Hu et al., 2020) and MMLU-ProX (Xuan et al., 2025) provide standardized cross-language comparison, no equivalent standardized benchmark currently exists specifically for evaluating LLM reliability in scientific literature analysis tasks, such as claim verification, citation accuracy, or methodological summarization, across languages. Existing evaluation frameworks for automated literature review, including reference-hallucination metrics and semantic-coverage measures, have to date been validated primarily on English-language corpora, and their applicability to non-English scientific text has not been systematically established.

### *5.5 Evaluation and Reporting Limitations*

Several of the studies synthesized in this review, including large-scale systematic reviews of LLM-assisted review automation, explicitly excluded non-English publications from their own inclusion criteria, meaning that the very evidence base used to characterize the state of LLM-assisted literature analysis is disproportionately drawn from English-language use cases and applications. This creates a structural risk that the field's understanding of LLM reliability in scientific contexts is itself linguistically biased, independent of the underlying capability of the models being studied.

## 6. Research Gaps and Future Directions

### *6.1 Development of Multilingual Scientific-Reliability Benchmarks*

A clear priority for future research is the development of standardized, expert-validated benchmarks that specifically evaluate LLM reliability on scientific literature analysis tasks across a typologically and resource-diverse set of languages, following the design principles established by general-purpose multilingual benchmarks such as XTREME and MMLU-ProX but extended to include claim verification, citation-grounding accuracy, and cross-lingual factual consistency specific to scientific domains such as biomedicine, materials science, and the social sciences.

### *6.2 Investment in Non-English Scientific Corpora and Annotation Infrastructure*

Given that corpus scarcity is repeatedly identified as a root constraint, future work should prioritize the systematic collection, curation, and annotation of non-English scientific literature, potentially through coordinated efforts analogous to the BigScience initiative that produced the ROOTS corpus underlying BLOOM (BigScience Workshop et al., 2022), but purpose-built for scientific and technical domains rather than general web text.

### *6.3 Disentangling Language-Resource and Domain-Specific Effects*

Future evaluation designs should explicitly separate language-resource effects from domain-specific effects, for example by evaluating the same scientific subdomain across languages of varying resource levels, and the same language across scientific subdomains of varying training-data density, in order to more precisely target mitigation strategies and to avoid conflating distinct sources of unreliability.

### *6.4 Cross-Lingual Verification and Grounding Architectures for Literature Synthesis*

Building on demonstrated but partial successes of retrieval augmentation and cross-lingual chain-of-thought prompting, future systems should explore tighter integration between multilingual retrieval infrastructure and literature-synthesis pipelines, including multilayered quality-control architectures similar to those reported by Wu et al. (2025) and the multi-agent screening frameworks exemplified by LatteReview, explicitly extended and validated for non-English source material rather than assumed to generalize from English-only validation.

### *6.5 Equity-Oriented Evaluation of Knowledge Propagation*

Because Qi et al. (2023) demonstrate that newly inserted factual knowledge in English propagates unevenly to other languages depending on pre-existing cross-lingual consistency, future research should examine the equity implications of this dynamic for global scientific communication: specifically, whether LLM-mediated access to recent scientific findings systematically disadvantages researchers and

practitioners querying in non-English languages, and what architectural or training interventions could most directly address this propagation gap.

### *6.6 Standardized Reporting of Language Coverage in LLM-Assisted Review Tools*

Finally, future systematic reviews and empirical evaluations of LLM-assisted literature analysis tools should explicitly report the language coverage of both their training and evaluation data, rather than implicitly generalizing findings derived from English-only pipelines. Establishing this as a reporting norm, analogous to existing expectations around dataset documentation, would substantially improve the field's ability to track progress on multilingual reliability over time.

## 7. Conclusion

The evidence reviewed in this paper indicates that large language models, despite substantial and well-documented progress in multilingual capability, remain measurably less reliable when applied to scientific literature analysis tasks in languages other than English. This unreliability manifests along several interrelated dimensions: declining accuracy on multilingual reasoning benchmarks as language-resource level decreases (Xuan et al., 2025), cross-lingual factual inconsistency in which the same query yields different answers depending on the language used (Qi et al., 2023), and persistent hallucination in automated literature synthesis pipelines even when tested primarily on English-language corpora (Tang et al., 2025; Wu et al., 2025). Current mitigation strategies, including retrieval augmentation, cross-lingual chain-of-thought prompting, targeted multilingual and domain-specific fine-tuning, and multilayered human-in-the-loop quality control, each demonstrate measurable but partial improvements, and none has yet been shown to close the reliability gap fully, particularly for scientific applications in lower-resource languages.

Because scientific knowledge production and consumption are inherently multilingual, and because LLMs are already being deployed at scale to assist with literature search, screening, and synthesis, closing this reliability gap is not merely a technical curiosity but a matter of equitable access to scientific knowledge. Future progress will likely depend on coordinated investment in non-English scientific corpora and annotation infrastructure, the development of standardized multilingual scientific-reliability benchmarks, and a disciplinary norm of transparent reporting regarding the language coverage of both the training data and the evaluation evidence underlying claims about LLM capability in scientific contexts.

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